

FORTIGATE NSE4

## Lab 2 — Configuring FortiGate

## 1. Topology

This lab extends Lab 1 by adding a second test client behind a Layer 2 switch. Two hosts on the same segment is what makes the VLAN segmentation question meaningful — with one client there is nothing to isolate.

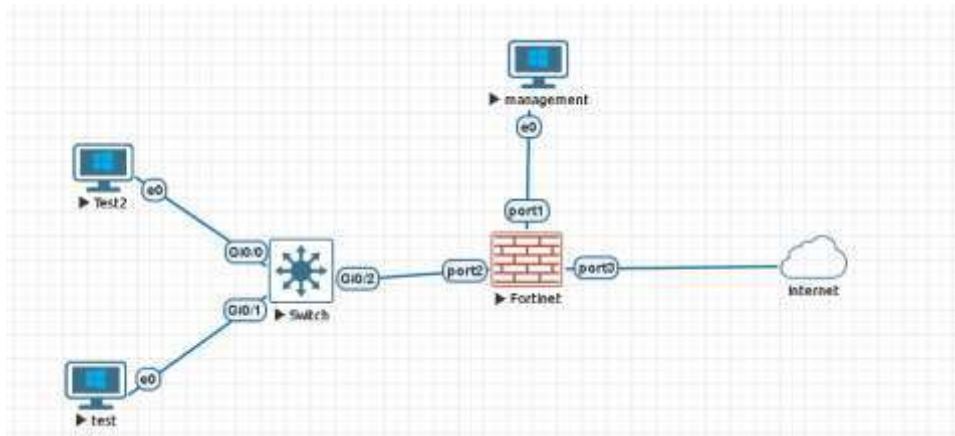


Figure 1 — Management PC, two test clients behind a switch, and the internet cloud

| Port  | Alias | Role       | Connected to                 |
|-------|-------|------------|------------------------------|
| port1 | MGMT  | Management | Management PC, 192.168.100.x |
| port2 | LAN   | Inside     | Switch → Test and Test2      |
| port3 | WAN   | Outside    | Internet cloud               |

## 2. LAN Interface Configuration

### ► 2.1 port2 as the LAN Gateway

port2 becomes the default gateway for the LAN segment, with a DHCP server so the test clients address themselves.

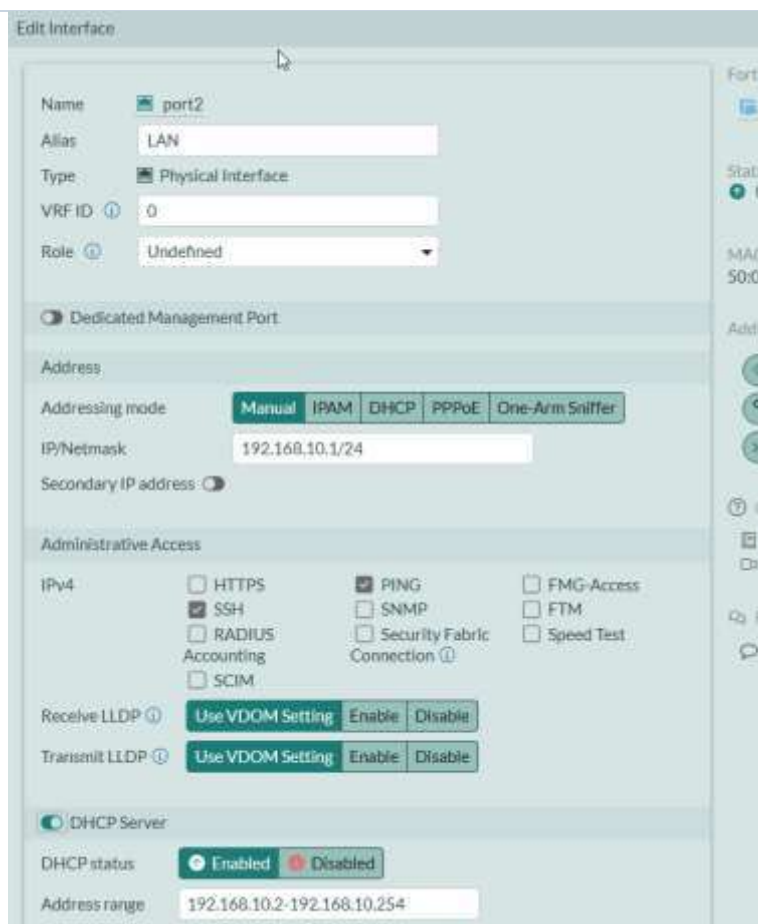


Figure 2 — port2 configured with a static address and DHCP enabled

| Setting               | Value               | Notes                                           |
|-----------------------|---------------------|-------------------------------------------------|
| Alias                 | LAN                 | Named for its role rather than its number       |
| Addressing mode       | Manual              | A gateway cannot take its own address from DHCP |
| IP / Netmask          | 192.168.10.1/24     | —                                               |
| Administrative access | PING, SSH           | No HTTPS — management belongs on port1          |
| DHCP server           | Enabled             | —                                               |
| DHCP range            | 192.168.10.2 - .254 | —                                               |

**Leaving HTTPS off this interface is the point of Lab 1 applied.** Management access lives on port1. **A host on the LAN segment should not be able to reach the firewall's own login page** — that separation is only worth anything if it is maintained as each new interface is added.

*Running DHCP on the firewall is acceptable in a lab or a small site. In a production industrial network, a dedicated DHCP server gives better lease control and a usable audit trail.*

## ► 2.2 Verifying DHCP

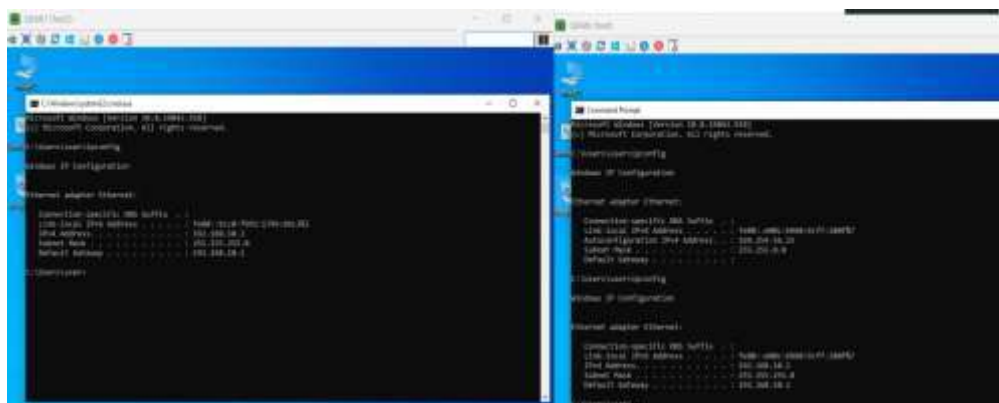


Figure 3 — Both test clients holding addresses from the FortiGate pool

Both machines received addresses in 192.168.10.x with the correct default gateway. This confirms the interface, the scope, and the switch path in one check.

## 3. VLAN Segmentation

### ► 3.1 The Problem

Both test clients sit on port2, so they share one broadcast domain and one DHCP pool. Isolating them into separate subnets with separate scopes requires Layer 2 separation first.

### ► 3.2 Why DHCP Forces the VLAN Question

**FortiGate binds a DHCP scope to an interface. A VLAN is a logical interface, so each VLAN gets its own scope — and without VLANs there is no way to run two scopes on one physical port.** The addressing requirement and the segmentation requirement turn out to be the same requirement.

- DHCP is assigned per interface
- A VLAN is a logical interface, also called a sub-interface
- No VLAN means no second DHCP scope on that port

### ► 3.3 Converting port2 to a Trunk

Before VLAN sub-interfaces can be created, the parent interface must give up its own addressing. The IP is removed and DHCP disabled, leaving port2 as a pure trunk.



Figure 4 — port2 with its address removed, carrying PING and SSH access only

### ► 3.4 Evaluation Licence Limitation

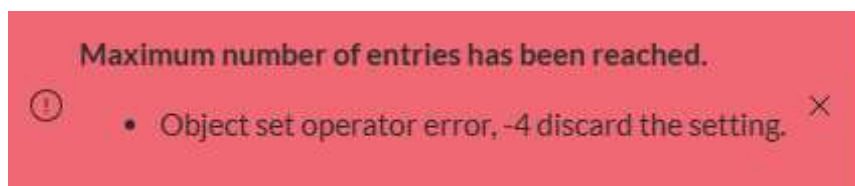


Figure 5 — The licensing error raised when a second VLAN is created on the same port

**The evaluation licence caps the number of interface entries, so creating two VLANs on one physical port fails** with a maximum-entries error. The configuration is correct; the licence is the constraint.

**Workaround used:** two separate physical interfaces, one per segment, with the separation performed at the switch rather than by router-on-a-stick. The outcome is the same isolation; only the mechanism differs.

### ► 3.5 Router-on-a-Stick on a Licensed Appliance

On a fully licensed FortiGate, VLAN sub-interfaces are created on the single physical port, each with its own VLAN ID, address and DHCP scope.

- Create new interfaces of type VLAN
- Assign the VLAN IDs — 10 and 20
- Bind each to the parent interface, port2
- Configure a separate DHCP pool per VLAN

### ► 3.6 Switch Configuration

```
! Trunk toward the FortiGate
interface GigabitEthernet0/2
  switchport mode trunk
  switchport trunk allowed vlan 10,20
!
! Access port for Test – VLAN 10
interface GigabitEthernet0/0
  switchport mode access
  switchport access vlan 10
!
! Access port for Test2 – VLAN 20
interface GigabitEthernet0/1
  switchport mode access
  switchport access vlan 20
```

**Restricting the allowed VLAN list is worth doing deliberately.** A trunk carries every VLAN by default. **Naming only 10 and 20 means a VLAN created later is not carried across the trunk unless it is added explicitly** — the trunk stops being an implicit path for whatever happens to exist on the switch.

### ► 3.7 Verification

```
diagnose ip dhcp lease-list
```

*Expected: 192.168.10.x leases on VLAN 10, 192.168.20.x on VLAN 20*

## 4. Static Routing

Figure 6 — Default route toward the upstream gateway via port3

| Field                   | Value           | Notes                                                           |
|-------------------------|-----------------|-----------------------------------------------------------------|
| Destination             | 0.0.0.0/0.0.0.0 | Default route — everything not matched by a more specific entry |
| Gateway address         | 192.168.1.1     | The upstream router                                             |
| Interface               | WAN — port3     | —                                                               |
| Administrative distance | 10              | See below                                                       |
| Status                  | Enabled         | —                                                               |

**Administrative distance decides between competing routes.** The default for a static route is 10 on FortiOS. A lower value is preferred, so **a second default route at a higher distance acts as a standby — used only when the primary is withdrawn**. That is the basis of simple WAN failover before SD-WAN is introduced.

## 5. Firewall Policy

A FortiGate forwards nothing without an explicit policy. **This is the fundamental difference from a traditional router, which forwards by default and filters by exception.**

Every traffic flow needs a matching policy, and anything unmatched hits the implicit deny at the bottom of the table.

### ► 5.1 LAN to Internet



Figure 7 — The outbound policy as it appears in the policy table

| Field               | Value                | Reasoning                                              |
|---------------------|----------------------|--------------------------------------------------------|
| Name                | LAN_TEST_to_Internet | Appears in the traffic log, so it should be readable   |
| Incoming / Outgoing | port2 → port3        | LAN toward WAN                                         |
| Source              | 192.168.10.0/24      | Scoped to the LAN rather than opened to all            |
| Destination         | all                  | Any internet destination                               |
| Service             | ALL                  | Narrow this in production to what is actually required |
| Action              | ACCEPT               | —                                                      |
| NAT                 | <b>Enabled</b>       | Required — see below                                   |
| Log allowed traffic | All sessions         | Without this the forward traffic log stays empty       |

NAT translates the private source addresses to the WAN interface address. **Without it the internal addresses are not routable upstream and no reply can find its way back** — which is the opposite of the inter-VLAN case, where NAT would destroy source attribution for no benefit.

## ► 5.2 Policy Processing Order

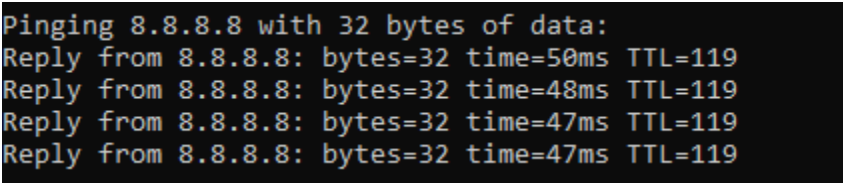
Policies are evaluated top to bottom and the first match wins. Everything below a matching rule is never considered, which makes order as significant as content.

- Specific rules — blocking an application or URL — go above the general allow
- The implicit deny sits at the bottom; unmatched traffic is dropped
- VLAN-specific policies need their own entry each — VLAN 10 to WAN and VLAN 20 to WAN are separate rules

**A broad accept placed above a specific deny silently disables the deny.** The policy list still shows both, and nothing reports the conflict — which is why order is checked as part of any policy review rather than assumed.

## 6. Verification

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```
Pinging 8.8.8.8 with 32 bytes of data:  
Reply from 8.8.8.8: bytes=32 time=50ms TTL=119  
Reply from 8.8.8.8: bytes=32 time=48ms TTL=119  
Reply from 8.8.8.8: bytes=32 time=47ms TTL=119  
Reply from 8.8.8.8: bytes=32 time=47ms TTL=119
```

*Figure 8 — End-to-end connectivity from the test client*

Successful replies confirm the whole path: client → port2 → the FortiGate's routing decision → port3 → source NAT → internet → return. **A failure at any one of those stages produces the same timeout, which is why the components are configured and checked in order rather than all at once.**

## 7. Summary

| Component          | Detail                                                                                          |
|--------------------|-------------------------------------------------------------------------------------------------|
| LAN interface      | port2 — 192.168.10.1/24 with DHCP 192.168.10.2–254                                              |
| WAN interface      | port3 — internet-facing                                                                         |
| Management         | port1 — kept separate; no HTTPS on the LAN interface                                            |
| DHCP principle     | One scope per interface; a VLAN is a logical interface, so VLANs are what allow multiple scopes |
| VLAN approach      | Router-on-a-stick where licensed; separate physical interfaces under the evaluation licence     |
| Licence constraint | Maximum interface entries reached when creating a second VLAN on one port                       |
| Static route       | 0.0.0.0/0 via 192.168.1.1 on port3, administrative distance 10                                  |
| Firewall policy    | LAN to WAN, ACCEPT, NAT enabled, all sessions logged                                            |
| Default behaviour  | Implicit deny — nothing is forwarded without a matching policy                                  |
| Verification       | Ping to a public address from the test client, 0% loss                                          |